Simulating the behavior of volatiles belonging to the C-O-H-S system in silicate melts under magmatic conditions with the software D-Compress

Alain BURGISSER[1,2*], Marina ALLETTI[3,4], Bruno SCAILLET[3,4],

[1] CNRS, ISTerre, F-73376 Le Bourget du Lac, France.

[2] Université de Savoie, ISTerre, F-73376 Le Bourget du Lac, France.

[3] CNRS, ISTO, F-45071 Orléans, France.

[4] Université d'Orléans, ISTO, F-45071 Orléans, France.

* Corresponding author: Phone: (+33) 479 758 780. Fax: (+33) 479 758 742. Email: alain.burgisser@univ-savoie.fr

Appendix B - Supplementary Information

B1. Sulfur solubility laws in basalts

B2. Creating a new solubility law for H_2O in phonolite

B1. Sulfur solubility laws in basalts

To calibrate D-Compress, we considered five potentially suitable experimental sets addressing S solubility in basalt in the pressure range of interest (≤3000 bar): Beermann et al. (2011), Botcharnikov et al. (2011), Moune et al. (2009), Lesne et al. (2011), and Lesne et al. (in review). Here we review each experimental set separately to select suitable runs for calibration. Except where indicated, fluid compositions were assessed by using the same relationships as described in the "Summary of the chemical model" section (Eqs. 1-7), which corresponds to using D-Compress in gas only mode with the SOH system. When melt H_2O contents were needed to determine f_{H2O}, D-Compress was used in the SOH mode with the relevant magma compositions to ensure internal consistency. The breakdown of our final selection (selected runs/total runs) is 24/29, 3/6, and 7/11 for the respective sets published in Lesne et al. (in review), Botcharnikov et al. (2011), and Beermann et al. (2011). In most cases, the largest source of propagated error affects the fugacities of SO_2 or H_2S and comes from uncertainties on melt water content (H_2O_{melt}). Figure B1 shows these selected data and the associated solubility laws. Note that data scatter precludes the distinction of the effect of temperature on S solubility, so unique regressions were done for both SO_2 and H_2S over the temperature range from 1050 to 1250 °C.

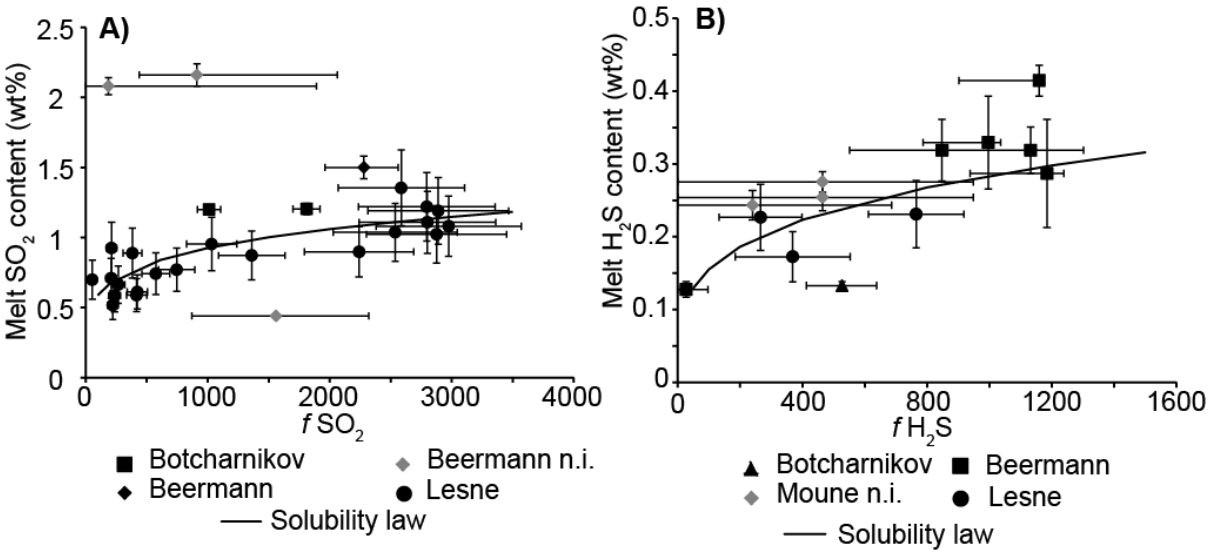

Figure B1: Summary of selected experimental data constraining S solubility in basalts. Solubility laws (new) are solid black curves. Data points shown in gray were not included (n.i.) in the regressions because of the large errors (see text). Errors are sometimes strongly asymmetric and thus may be displayed in only one direction. References are Beermann et al. (2011), Botcharnikov et al. (2011), Lesne et al. (in review), and Moune et al. (2009). A) Oxidized conditions where SO_2 dominates. A) Reduced conditions where H_2S dominates.

The first dataset we considered is that of Lesne et al. (2011), which simulates a closed-system decompression with volatiles from the CSOHCl set. Constrained variables for each experiment are P, T, the redox state, and the melt amounts of H_2O, CO_2, and S. Leaving S aside because we want to use it as calibration data, and Cl under the (unproven) assumption that its presence does not affect S and CO_2 solubilities, all other variables but the redox state have negligible or small enough uncertainties to allow us to include this set in our solubility law. The redox state is given at a precision of ±0.86 log unit, but we found that this level of precision is most likely underestimated. For all charges, the melt S content is constant above 50 to 100 MPa (see their Fig. 2d). If globally oxidized conditions are assumed (i.e., above NNO), that means that f_{SO2} is constant for all charges having total pressures

between 100 and 400 MPa. A coarse estimate for this constant f_{SO2} is ~1000 bar because at 1150°C and 100 MPa, γ_{SO2} is close to ideal at 1.15, so that to first order f_{SO2}=P. Taking for example the 300 MPa charge St8.1.A4, this means the redox state has to be <NNO+0.2, which is far below the NNO+1.7±0.86 given by the authors. This is true of all the runs, with the 2–3-orders-of-magnitude uncertainty on the redox state causing propagated errors on f_{SO2} of >300%. The redox state thus cannot be used as an input variable and this dataset has to be left aside. Note that if the conditions are as reduced as we predict (i.e., at or below NNO), the dissolved S is controlled by the combined effects of f_{SO2} and f_{H2S}, making the dataset unsuitable for calibration of D-Compress even if the redox state were known with sufficient precision. This internal inconsistency explains the difference between D-Compress outputs and the outputs of the model SolEx (Witham et al., 2012), which is based on the Lesne et al. (2011) dataset.

The difference between D-Compress and SolEx can be further illustrated by using P, T, and the melt amounts of H_2O, CO_2, and S as input data in D-Compress to calculate the redox state of each run. Figure B2 shows the redox state given by Lesne et al. (2011) and the estimates given by D-Compress with the two H_2O and CO_2 solubility laws available in the software(these laws are embedded in D-Compress through the toggle "T-X dependent solubilities" in the "Advanced Parameters" window, see the User Manual for more details). Depending on the solubility law chosen, some low-pressure runs cannot be replicated because they contain more dissolved volatiles than that allowed by the solubility laws. Overall, D-Compress predicts that the run conditions were 4 to 5 log units more reduced than by Lesne et al. (2011). Exploring the reasons behind such a difference is outside the scope of the present manuscript.

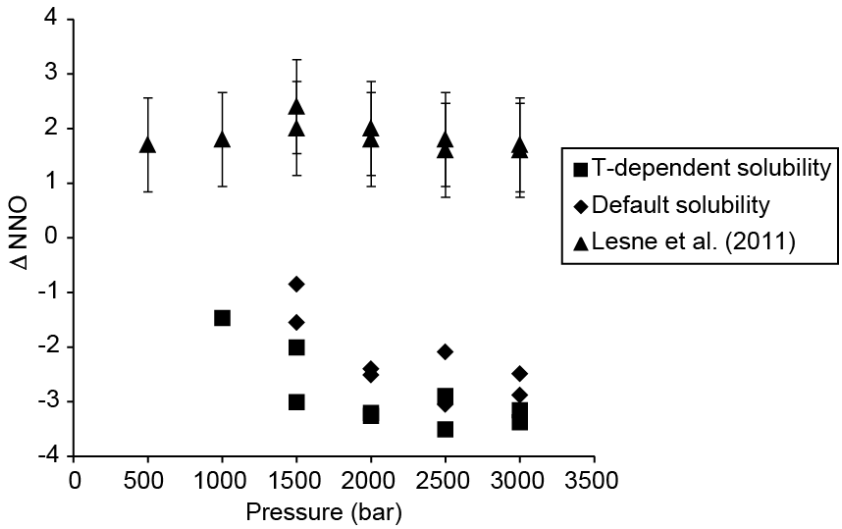

Figure B2: Redox conditions as a function of pressure in the experimental series St8.1 of Lesne et al. (2011). Triangles mark redox values calculated by Lesne et al. (2011) from measured Fe2+/Fe3+ ratios in the glass. Squares mark redox values using D-Compress and the T-dependent, H_2O and CO_2 solubility laws from Iacono Marziano et al. (2011). Diamonds mark redox values using D-Compress and the default, H_2O and CO_2 solubility laws using data from Lesne et al. (2011a, 2011b) (Table 1).

The second dataset is that of Beermann et al. (2011) who studied the SOH system in basalts. Keeping S melt contents for calibration, the remaining variables are P, T, redox state, and H_2O melt content (H_2O_{melt}). The redox state is obtained by using the measured S speciation in the final glasses. One H_2O-only run suggests that for this set the Iacono Marziano solubility law for H_2O and CO_2 is slightly more appropriate than that by default. The redox state of one sample (0_15_2 red) can also be calculated with D-Compress thanks to the measured values of f_{H2} and H_2O_{melt}. These two methods

yield redox states within 0.56 log unit of each other, which is a good agreement and leads us to accept the f_{O2} given for the other samples. The authors report f_{S2} following Liu et al. (2007) for 7 sulfide-bearing samples. The f_{S2} values obtained by D-Compress using P, T, and redox state as input parameters are ~3 orders of magnitude above the reported f_{S2}. This significant mismatch can be understood by using D-Compress to calculate instead the redox state from P, T, and f_{S2}. Figure B3 shows that there are two possible redox states that explain the reported f_{S2} values because the SOH system with 6 species is of quadratic nature (Burgisser and Scaillet, 2007). These values frame the 1:1 line and are thus not diagnostic of an internal inconsistency within the dataset. In the absence of other constraints, we decided that the six reduced runs from this set are suitable to be included in our H_2S solubility law. We suspect that the multiple stable redox states can explain redox oddities such as sample S2, which is classified as reduced by the authors because it contains PdFeS and PdS but has a reported redox state of NNO+0.7 with SO_2 as a dominant gas species (f_{SO2}= 1560 bar, f_{H2S}=104 bar, and f_{S2}=72 bar).

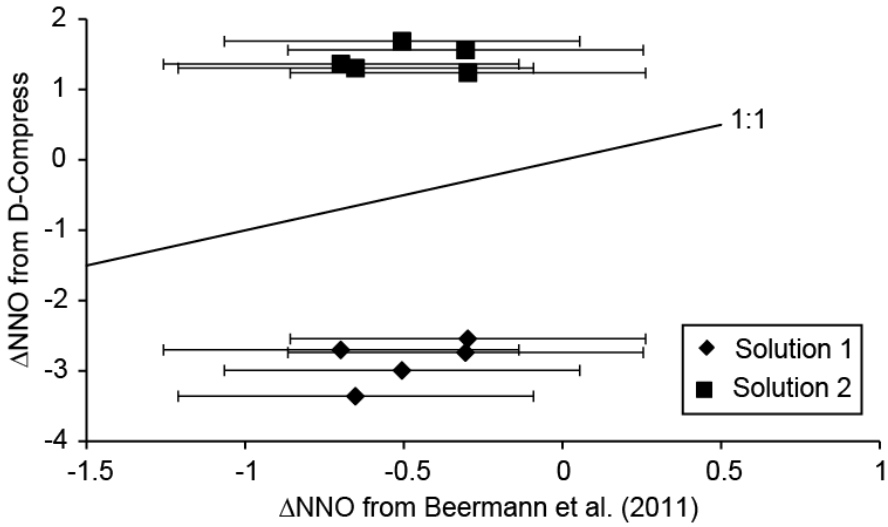

Figure B3: Redox states given by D-Compress as a function of the redox state given in Beermann et al. (2011) using S speciation in melt. Each input values of P, T, and f_{S2} in D-Compress yield two possible redox states, which correspond to two combinations of f_{H2S} and f_{SO2}.

We found that only one (S26) of the oxidized runs from the Beermann et al. (2011) set is suitable for our SO_2 solubility law. This is because one sample (0_152 ox) contains more water than the solubility law allows for, which precludes any calculation, and because the two other runs (S23 and S24) have uncertainties on H_2O_{melt} that yield large errors (50 to >>100%) on f_{SO2}. The large errors of these two runs and the fact that they have melt S content far from the trend defined by the other data points (Fig. B1) led us to discard them, which is perhaps linked to compositional shifts caused by the interaction between the experimental melts and the olivine containers.

The next dataset is that of Moune et al. (2009), who studied the SOH system in basalts. Keeping S melt contents for calibration, the remaining variables are P, T, redox state, and H_2O_{melt}. The redox state reported by the authors is obtained by using the f_{H2} imposed in the vessel and H_2O_{melt}. There are two issues with the internal consistency of this set. The first is linked to H_2O_{melt}. The only S-free, H_2O-staturated run (triangle in Fig. B4) has more dissolved water than all the other runs and than the two D-compress solubility laws would allow for. Moune et al. (2009) give no satisfactory explanation of why this only run would have an incorrect H_2O_{melt} measurement, so this casts a doubt on all the other reported H_2O_{melt} values.

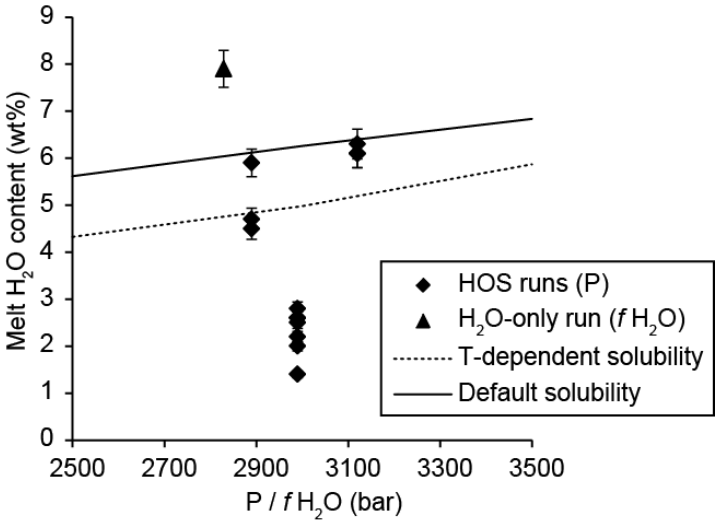

Figure B4: Melt H_2O content as a function of total pressure or f_{H2O} for data from Moune et al. (2009). The triangle represents run B1 with f_{H2O} calculated using the given P, T, and f_{O2} and the O-H system in D-Compress. Diamonds represent the other basalt runs of Moune et al. (2009) as a function of P. Assuming that f_{H2O}=P yields approximate maximum H_2O_{melt} values quickly indicate the most appropriate H_2O solubility law. The dotted curve is the T-dependent H_2O solubility law from Iacono Marziano et al. (2011) and the solid curve is the default H_2O solubility law using data from Lesne et al. (2011a, 2011b) (Table 1).

The second issue is the redox estimates. The authors report f_{S2} following Liu et al. (2007) for pyrrhotite-bearing samples. We determined f_{S2} using P, T, redox state, and H_2O_{melt} as initial parameters in D-Compress. We used the solubility law by default for H_2O and CO_2 as it ensures that no run has more water than that predicted for a melt statured with a pure H_2O fluid (Fig. B4). Figure B5 shows that only three samples have matching f_{S2} (diamonds). Imposing instead that the f_{S2} of both methods should be identical allows us to calculate a unique redox state that resolves this discrepancy. The triangles in Fig. B5 suggest that unique f_{S2} can be obtained if run conditions were oxidized, but in a range of values (NNO+0.6 to NNO+1.7) where the Liu et al. (2007) relationship is not calibrated and is thus not necessarily applicable. Overall, the many orders of magnitude between these redox and f_{S2} estimates preclude further interpretations.

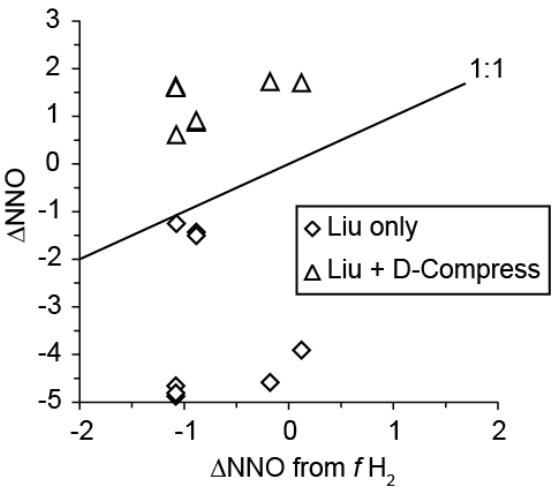

Figure B5: Redox states as calculated by two methods as a function of that given in Moune et al. (2009) from the experimentally imposed f_{H2}. Diamonds represent f_{S2}-constrained redox states derived from Liu et al. (2007) only and triangles represent f_{S2}-constrained redox states derived from both Liu et al. (2007) and D-Compress.

Considering these two issues, we decided that only the three runs with matching f_{S2} values between D-Compress and given by Moune et al. (2009) could be considered to calibrate D-Compress. Unfortunately, these runs are also the ones with the highest H_2O_{melt} (Fig. B4). As a result, the higher bounds of H_2O_{melt} given by the error margins are systematically above that of the solubility law by default. This yield uncertainties on f_{H2S} (these are reduced conditions) so large that including these three samples does not add more information to our solubility law (gray diamonds in Fig. B1). We thus left all runs from Moune et al. (2009) aside.

The next dataset is that of Botcharnikov et al. (2011), who studied the SOH system in basalts. Keeping S melt contents for calibration, the remaining variables are P, T, redox state, and H_2O_{melt}. The authors report the redox state using the f_{H2} imposed in the vessel and H_2O_{melt}. They report f_{S2} following Toulmin and Barton (1964) for two sulphide-bearing samples. To take into account that the sulfides might have been liquid at run conditions, we also calculated f_{S2} using Liu et al. (2007).The f_{S2} values obtained by D-Compress using P, T, and redox state as input parameters are consistent with those using both sulfide-based relationships (Fig. B6), which led us to give further consideration to all basaltic samples except SB41 as it is graphite-saturated. Among these 6 samples, two are oxidized enough to be included in our SO_2 solubility law (SB5 and SB15) and one is reduced enough to be included in our H_2S solubility law (SB36). The three others were left aside as their measured S melt contents are a complex function of both S species.

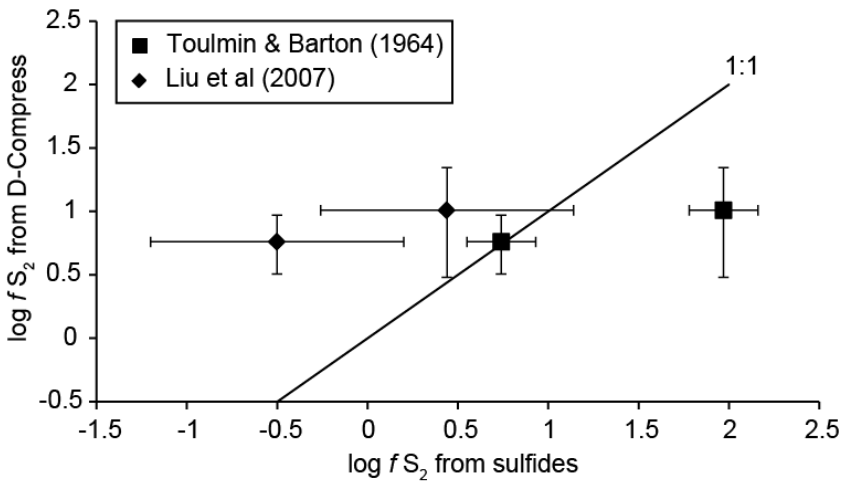

Figure B6: Fugacities of S_2 as calculated by two methods (squares and diamonds) as a function of those obtained by D-Compress.

The final dataset is that of Lesne et al. (in review), who studied the SOH system in basalts. Five runs were performed at 1050°C and had high H_2O_{melt}, which resulted in computed f_{H2O} higher than that of pure H_2O at the corresponding P and T. These five runs were left aside because it was not possible to derive fugacities of S-bearing species (Lesne et al., in review). As a result, 21 runs were included in the SO_2 solubility law and 3 runs were included in the H_2S solubility law (Fig. B1).

B2. Creating a new solubility law for H_2O in phonolite

The temperature-dependent solubility law for H_2O in phonolitic melts was found by using 116 out of 129 experiments from Carroll and Blank (1997), Larsen and Gardner (2004), Iacono Marziano et al. (2007), Schmidt and Behrens (2008), and Burgisser et al. (2012). When the data from all these sources are grouped in three temperature groups (825-900 °C, 950-1050 °C, and 1150-1200 °C), they

define clear trends of melt H_2O content as a function of f_{H2O}. One exception is a series at 850 °C from Schmidt and Behrens (2008), which defines a trend falling between the low and the mid-temperatures group. Removing this series from the low-temperature group and treating it apart yields four well-defined solubility laws of the form $a \times f_{H2O}^{\ b}$ (Fig. B7). When the coefficients *a* and *b* are plotted as a function of temperature (Fig. B8), the two coefficients for the 850 °C series from Schmidt and Behrens (2008) do not follow the trends defined by all the other data. This motivated us to exclude it from the temperature-dependent solubility law built into D-Compress (Table 1), which results from the quadratic interpolation of the coefficients shown on Fig B8.

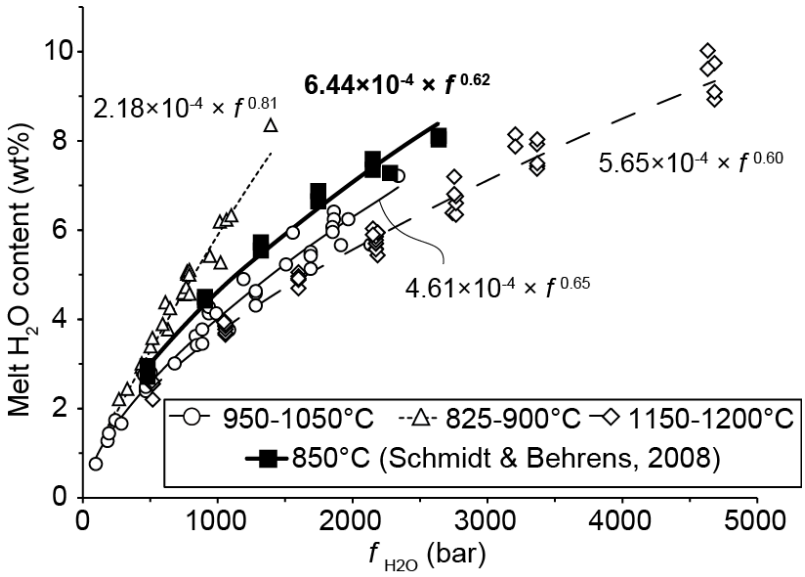

Figure B7: Melt H_2O content as a function of f_{H2O} for 129 experiments on phonolite. Four power laws are derived from grouping the data into groups (circles, triangles, diamonds, and squares). The 13 experiments from a series at 850 °C (squares) define a singular trend that is used as an example to build a new solubility law.

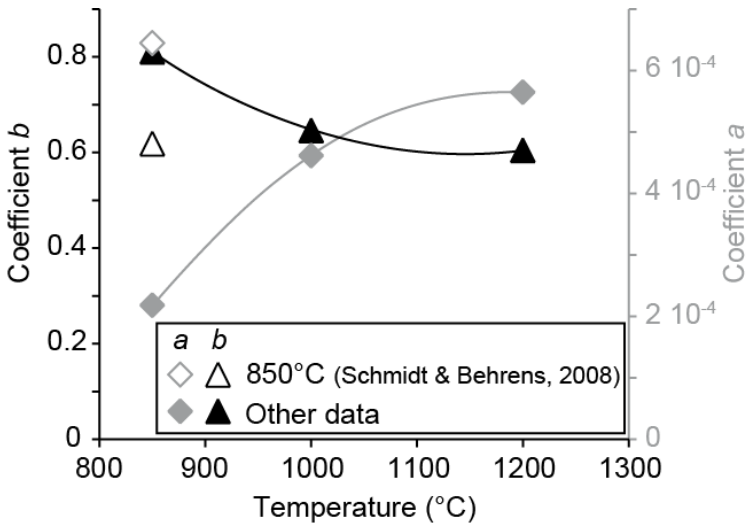

Figure B8: Solubility coefficients for H_2O as a function of temperature. Coefficients from the 850°C series of Schmidt and Behrens (2008) (open symbols) have values distinct from the trend defined by the other data (closed symbols) and are used to build a new solubility law. Curves are quadratic fits of the other data that link temperature and coefficient values.

Leaving aside the reasons (such as compositional effects) behind the peculiar behavior of this series, one can use it to build a new H_2O solubility law valid at 850 °C and for melt compositions

MBP and LSPII (Schmidt and Behrens, 2008; Table 2 of the user manual of D-Compress in Appendix A). Sometimes authors report total pressures instead of fugacities. It is possible to convert total pressure into f_{H2O} with D-Compress by selecting to O-H system and entering the redox state, temperature, and pressure of each experimental run and click on Compute initial conditions. Since both melt H_2O content and f_{H2O} are given in Schmidt and Behrens (2008), building Fig. B7 is straightforward. Since four significant digits are generally necessary to accurately represent the power law, the values a_{H2O}=6.4450×10^{-4} and b_{H2O}= 0.6182 can be entered into D-Compress (User-defined melt composition in the Advanced parameters window) and the new law can finally be saved in an .ini file using the Save settings button on the main window.

B3. Comparison between experimental and calculated isobaric melt volatile contents

It is customary to present the predictive capability of a model with graphs of measured vs. calculated values (Fig. B9). The solubility laws used in D-Compress predict calculated values with uncertainties of three distinct origins. The first source is melt composition. Data in Table 2 show that solubility laws are often valid for a range of melt composition, but these ranges are quite narrow. There is no quantification of the uncertainties associated with melt composition in the solubility laws of the form $a_i \times f_i^{bi}$ because, unlike the H_2O-CO_2 laws of Iacono Marziano et al. (2012), they do not explicitly depend on melt composition. The second cause is the intrinsic uncertainties in the concentration vs. fugacity regressions, which are quantified by the extreme solubility laws (Figs 9-10, B1). Intrinsic uncertainties are reported as single gray horizontal bars for each species and each composition on Fig. B9. The last cause is the uncertainties in the experimental input parameters used to perform the calculation: temperature, pressure, and, for species other than water: redox state, melt H_2O content or f_{H2O}, and CO_2 melt content. This extrinsic source of uncertainty is displayed as horizontal error bars for every data point on Fig. B9.

Figure B9 shows that calibration of H_2O has low intrinsic and extrinsic uncertainties. Calibration of CO_2 shows large measurement uncertainties but low intrinsic errors. Finally, S calibration has large intrinsic uncertainties, which cause considerable data scatter (Figs. B9C, B9F, and B9I). This data scatter can be reduced a little by making H_2S solubility coefficients dependent on temperature and restricting the calibration range from 800 – 1000 °C to 800– 900 °C (Burgisser et al., 2008). The predictive capability of the solubility model is shown by plotting other data on Fig. B9. Poor prediction caused by the influence of melt composition is shown in Figs. B9A, B9E, and B9G. This situation can be addressed by defining new solubility coefficients. The case B9G is treated in this way in the Section B2 of this Appendix. Figures B9B and B9D show a good level of prediction because the test data plot within the scatter defined by the calibration data. Fig B9C and B9I show that treating reduced and oxidized melts separately for S solubility yields good predictions for mixed fluids (i.e., with comparable proportions of reduced and oxidized S), even if the phonolite data suffer from very large extrinsic uncertainties.

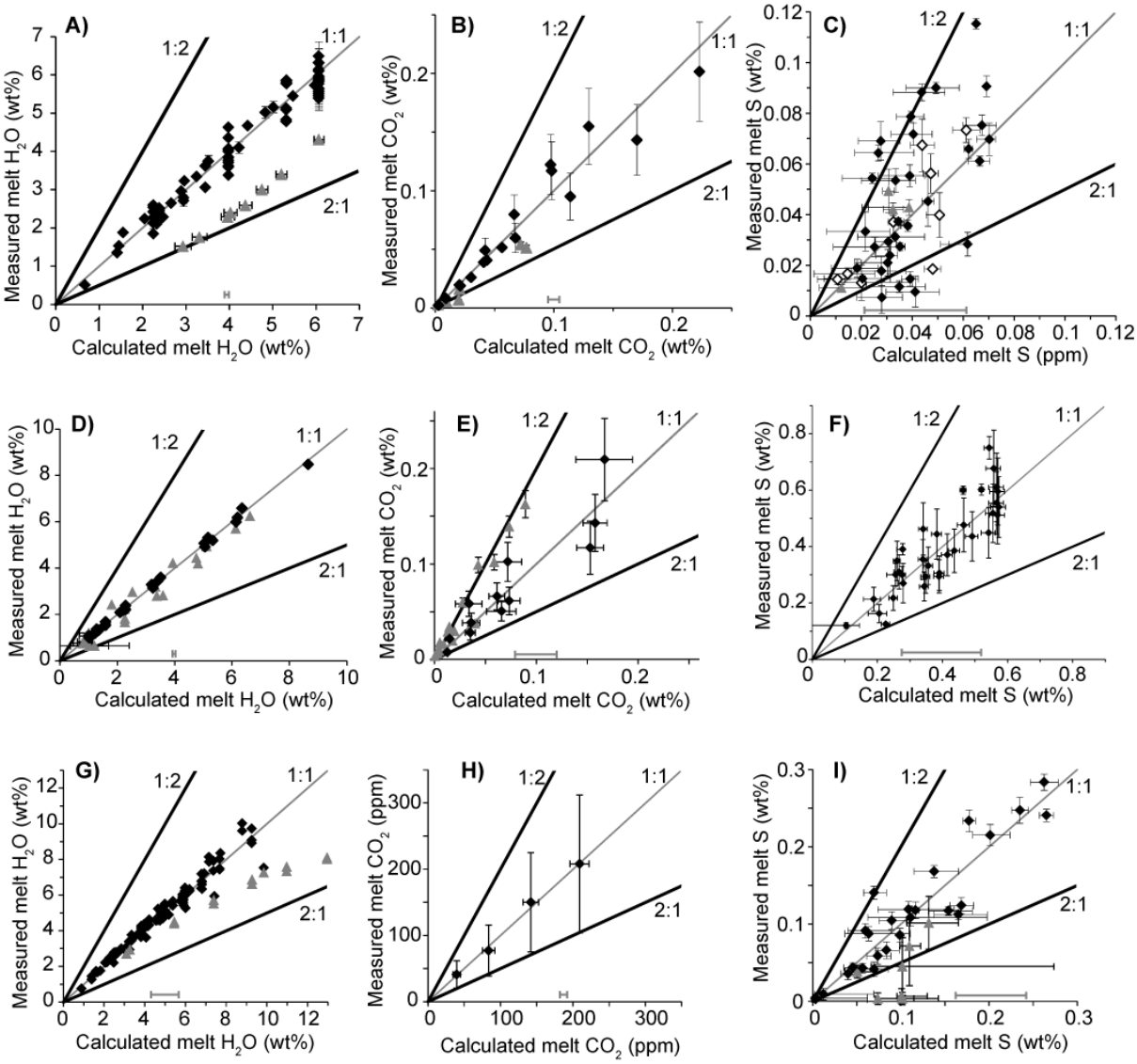

Figure B9: Measured vs. calculated dissolved species concentrations. Diamonds are calibration data (Tables 1-2) and triangles are other data. Horizontal gray error bars located near the middle point on the x-axis are representative intrinsic uncertainties resulting from extreme solubility law fittings. Black horizontal error bars attached to each experimental point (except when smaller than the respective symbol) result from uncertainties in the input data. A) H_2O in rhyolitic melt. Other data are from Tamic et al. (2001). B) CO_2 in rhyolitic melt. Other data are from Mangan and Sisson (2000). C) S in rhyolitic melt. Filled diamonds are H_2S calibration data and open diamonds are SO_2 calibration data. Other data are mixed fluid (1<NNO<1.6) from Clemente et al. (2004). D) H_2O in basaltic melt. Other data are from Shishkina et al. (2010). E) CO_2 in basaltic melt. Other data are from Shishkina et al. (2010). F) S in basaltic melt. G) H_2O in phonolitic melt. Other data are from Schmidt and Behrens (2008). H) CO_2 in phonolitic melt. A 50% error on the absolute value of measured melt CO_2 contents is assumed because of the uncertainty in extinction coefficients of the carbonate doublet (Iacono Marziano, 2004). I) S in phonolitic melt. Other data are mixed fluid (-1<NNO<1) from Moncrieff (1999).

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